# Quick Commerce Inventory Optimization & Demand Intelligence System

Retail Analytics | Supply Chain Analytics | Quick Commerce Operations

Tools Used: Python | SQL | Power BI

## Executive Summary

This project focused on improving visibility into inventory performance, operational demand behavior, stockout risk, and replenishment planning across quick commerce stores and product categories.

Using Python, SQL, and Power BI, the project transformed raw operational data into actionable business insights supporting forecasting analysis, inventory optimization, and operational decision-making.

## Business Problem

Quick commerce businesses often face operational challenges due to unstable demand fluctuations, inventory imbalance, frequent stockouts, and weak replenishment visibility.

The objective of this project was to improve inventory planning visibility, identify operational risks, monitor forecasting performance, and support smarter replenishment decisions using data analytics.

## Tools & Technologies Used

Python
Pandas
NumPy
MySQL
Power BI
DAX
Excel

## Dataset Overview

The dataset was created using simulated operational retail data combined with weather-based demand influence factors.

The analysis included:
• 3 operational stores
• 18 operational products
• Multiple retail categories
• Forecasting metrics
• Inventory KPIs
• Weather-based operational data

Figure 1: Operational retail dataset preview

| | date | store_id | product_id | product_name | category | ... | inventory_qty | fulfilled_qty | stockout_flag | temperature | weather_type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2025-01-01 | 101 | P01 | Milk | Dairy | ... | 55 | 55 | 1 | 25.828065 | Clear |
| 1 | 2025-01-01 | 101 | P02 | Curd | Dairy | ... | 54 | 51 | 0 | 25.828065 | Clear |
| 2 | 2025-01-01 | 101 | P03 | Butter | Dairy | ... | 49 | 41 | 0 | 25.828065 | Clear |
| 3 | 2025-01-01 | 101 | P04 | Paneer | Dairy | ... | 43 | 35 | 0 | 25.828065 | Clear |
| 4 | 2025-01-01 | 101 | P05 | Bread | Bakery | ... | 58 | 52 | 0 | 25.828065 | Clear |
| 5 | 2025-01-01 | 101 | P06 | Buns | Bakery | ... | 30 | 30 | 1 | 25.828065 | Clear |
| 6 | 2025-01-01 | 101 | P07 | Cake | Bakery | ... | 28 | 28 | 1 | 25.828065 | Clear |
| 7 | 2025-01-01 | 101 | P08 | Chips | Snacks | ... | 59 | 54 | 0 | 25.828065 | Clear |
| 8 | 2025-01-01 | 101 | P09 | Biscuits | Snacks | ... | 53 | 47 | 0 | 25.828065 | Clear |
| 9 | 2025-01-01 | 101 | P10 | Namkeen | Snacks | ... | 28 | 27 | 0 | 25.828065 | Clear |
| 10 | 2025-01-01 | 101 | P11 | Coke | Beverages | ... | 36 | 36 | 0 | 25.828065 | Clear |
| 11 | 2025-01-01 | 101 | P12 | Juice | Beverages | ... | 26 | 26 | 1 | 25.828065 | Clear |
| 12 | 2025-01-01 | 101 | P13 | Soda | Beverages | ... | 17 | 17 | 1 | 25.828065 | Clear |
| 13 | 2025-01-01 | 101 | P14 | Energy Drink | Beverages | ... | 30 | 25 | 0 | 25.828065 | Clear |
| 14 | 2025-01-01 | 101 | P15 | Eggs | Essentials | ... | 44 | 44 | 1 | 25.828065 | Clear |
| 15 | 2025-01-01 | 101 | P16 | Oil | Essentials | ... | 14 | 14 | 1 | 25.828065 | Clear |
| 16 | 2025-01-01 | 101 | P17 | Rice | Essentials | ... | 46 | 46 | 1 | 25.828065 | Clear |
| 17 | 2025-01-01 | 101 | P18 | Atta | Essentials | ... | 29 | 29 | 1 | 25.828065 | Clear |
| 18 | 2025-01-01 | 102 | P01 | Milk | Dairy | ... | 54 | 50 | 0 | 25.828065 | Clear |
| 19 | 2025-01-01 | 102 | P02 | Curd | Dairy | ... | 44 | 44 | 1 | 25.828065 | Clear |

## Data Cleaning & Feature Engineering

Data preprocessing and feature engineering were performed using Python to improve analytical reliability and forecasting quality.

The following transformations were applied:

- Fill rate calculation
- Demand-inventory gap generation
- Weekend demand identification
- Stockout validation
- Forecasting feature creation

Figure 2: Operational feature engineering and KPI creation workflow

```python
# Feature Engineering
df['date'] = pd.to_datetime(
    df['date'],
    format='mixed',
    dayfirst=True
)
df['day_of_week'] = df['date'].dt.day_name()
df['month'] = df['date'].dt.month
df['weekend'] = df['date'].dt.weekday >= 5

# DEMAND-INVENTORY GAP
df['gap'] = df['demand_qty'] - df['inventory_qty']

# Fill rate
df['fill_rate'] = df['fulfilled_qty'] / df['demand_qty']

# STOCKOUT VALIDATION
df['calc_stockout'] = (df['demand_qty'] > df['inventory_qty']).astype(int)
```

Figure 3: Cleaned and transformed operational dataset

| | date | store_id | product_id | product_name | category | demand_qty | ... | day_of_week | month | weekend | gap | fill_rate | calc_stockout |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2025-01-01 | 101 | P01 | Milk | Dairy | 63 | ... | Wednesday | 1 | False | 8 | 0.873016 | 1 |
| 1 | 2025-01-01 | 101 | P02 | Curd | Dairy | 51 | ... | Wednesday | 1 | False | -3 | 1.000000 | 0 |
| 2 | 2025-01-01 | 101 | P03 | Butter | Dairy | 41 | ... | Wednesday | 1 | False | -8 | 1.000000 | 0 |
| 3 | 2025-01-01 | 101 | P04 | Paneer | Dairy | 35 | ... | Wednesday | 1 | False | -8 | 1.000000 | 0 |
| 4 | 2025-01-01 | 101 | P05 | Bread | Bakery | 52 | ... | Wednesday | 1 | False | -6 | 1.000000 | 0 |
| 5 | 2025-01-01 | 101 | P06 | Buns | Bakery | 44 | ... | Wednesday | 1 | False | 14 | 0.681818 | 1 |
| 6 | 2025-01-01 | 101 | P07 | Cake | Bakery | 32 | ... | Wednesday | 1 | False | 4 | 0.875000 | 1 |
| 7 | 2025-01-01 | 101 | P08 | Chips | Snacks | 54 | ... | Wednesday | 1 | False | -5 | 1.000000 | 0 |
| 8 | 2025-01-01 | 101 | P09 | Biscuits | Snacks | 47 | ... | Wednesday | 1 | False | -6 | 1.000000 | 0 |
| 9 | 2025-01-01 | 101 | P10 | Namkeen | Snacks | 27 | ... | Wednesday | 1 | False | -1 | 1.000000 | 0 |
| 10 | 2025-01-01 | 101 | P11 | Coke | Beverages | 36 | ... | Wednesday | 1 | False | 0 | 1.000000 | 0 |
| 11 | 2025-01-01 | 101 | P12 | Juice | Beverages | 30 | ... | Wednesday | 1 | False | 4 | 0.866667 | 1 |
| 12 | 2025-01-01 | 101 | P13 | Soda | Beverages | 30 | ... | Wednesday | 1 | False | 13 | 0.566667 | 1 |
| 13 | 2025-01-01 | 101 | P14 | Energy Drink | Beverages | 25 | ... | Wednesday | 1 | False | -5 | 1.000000 | 0 |
| 14 | 2025-01-01 | 101 | P15 | Eggs | Essentials | 53 | ... | Wednesday | 1 | False | 9 | 0.830189 | 1 |
| 15 | 2025-01-01 | 101 | P16 | Oil | Essentials | 26 | ... | Wednesday | 1 | False | 12 | 0.538462 | 1 |
| 16 | 2025-01-01 | 101 | P17 | Rice | Essentials | 48 | ... | Wednesday | 1 | False | 2 | 0.958333 | 1 |
| 17 | 2025-01-01 | 101 | P18 | Atta | Essentials | 36 | ... | Wednesday | 1 | False | 7 | 0.805556 | 1 |
| 18 | 2025-01-01 | 102 | P01 | Milk | Dairy | 50 | ... | Wednesday | 1 | False | -4 | 1.000000 | 0 |
| 19 | 2025-01-01 | 102 | P02 | Curd | Dairy | 48 | ... | Wednesday | 1 | False | 4 | 0.916667 | 1 |

## Forecasting & Predictive Analytics

A rolling 7-day forecasting model was implemented using historical product-store demand behavior to estimate future operational demand trends.

Figure 4: Product-store level rolling demand forecasting

```
# FORECASTING

df = df.sort_values(by=['store_id', 'product_name', 'date'])

df['forecast'] = (
    df.groupby(['store_id', 'product_name'])['demand_qty']
    .transform(lambda x: x.rolling(7).mean().shift(1))
)
```

Figure 5: Product-store level rolling demand forecasting output

| | date | store_id | product_name | demand_qty | forecast |
|---|---|---|---|---|---|
| 378 | 2025-01-08 | 101 | Atta | 42 | 42.571429 |
| 379 | 2025-01-08 | 101 | Biscuits | 46 | 38.428571 |
| 380 | 2025-01-08 | 101 | Bread | 50 | 59.285714 |
| 381 | 2025-01-08 | 101 | Buns | 45 | 47.571429 |
| 382 | 2025-01-08 | 101 | Butter | 35 | 42.571429 |
| 383 | 2025-01-08 | 101 | Cake | 28 | 30.857143 |
| 384 | 2025-01-08 | 101 | Chips | 37 | 49.285714 |
| 385 | 2025-01-08 | 101 | Coke | 32 | 32.571429 |
| 386 | 2025-01-08 | 101 | Curd | 57 | 49.428571 |
| 387 | 2025-01-08 | 101 | Eggs | 54 | 54.000000 |
| 388 | 2025-01-08 | 101 | Energy Drink | 20 | 22.142857 |
| 389 | 2025-01-08 | 101 | Juice | 37 | 39.714286 |
| 390 | 2025-01-08 | 101 | Milk | 68 | 63.571429 |
| 391 | 2025-01-08 | 101 | Namkeen | 28 | 34.428571 |
| 392 | 2025-01-08 | 101 | Oil | 35 | 29.571429 |

Key Forecasting Findings:
Forecast Accuracy: 84.38%
Average Forecast: 41.92
Demand Variability: 6.08
High-Risk SKUs: 18

## Inventory & Stockout Analysis

Inventory analysis revealed elevated stockout exposure and operational replenishment pressure across multiple product categories.

Key Operational KPI Findings
Total Demand: 409,879
Fulfilled Quantity: 363,823
Fill Rate: 87.60%
Stockout Rate: 59.54%
Lost Sales: 28,349

Figure 6: Operational KPI performance overview


## Category & Product Risk Analysis

Category-level analysis identified inventory pressure across Essentials, Beverages, and Dairy categories.
The analysis also identified operationally risky products experiencing repeated stockout occurrences, including Paneer, Oil, Juice, Rice, and Buns.

Figure 7: Category-level inventory risk analysis

Figure 8: High-risk SKU operational exposure

| product_name | stockouts |
|---|---|
| Paneer | 347 |
| Oil | 342 |
| Juice | 339 |
| Rice | 337 |
| Buns | 332 |
| Eggs | 328 |
| Soda | 325 |
| Cake | 323 |
| Energy Drink | 323 |
| Milk | 323 |
| Chips | 322 |
| Namkeen | 321 |
| Atta | 317 |
| Bread | 315 |
| Biscuits | 314 |
| Coke | 309 |
| Butter | 307 |
| Curd | 295 |

## Store Performance & Forecast Error Analysis

Store-level operational analysis showed relatively balanced performance across all stores.

Forecast error analysis highlighted higher variability among fast-moving products such as Chips, Biscuits, and Namkeen.

Figure 9: Store-level operational performance comparison

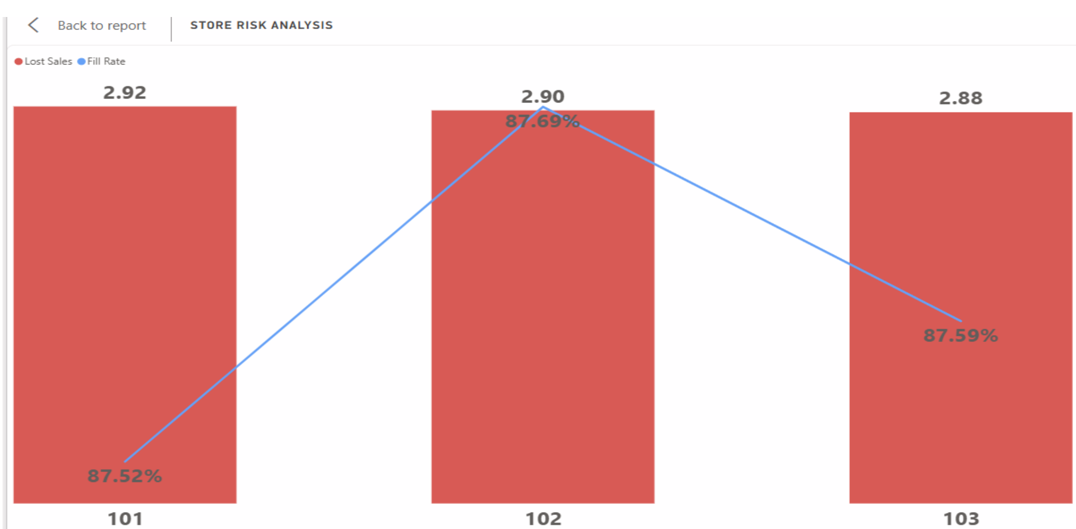
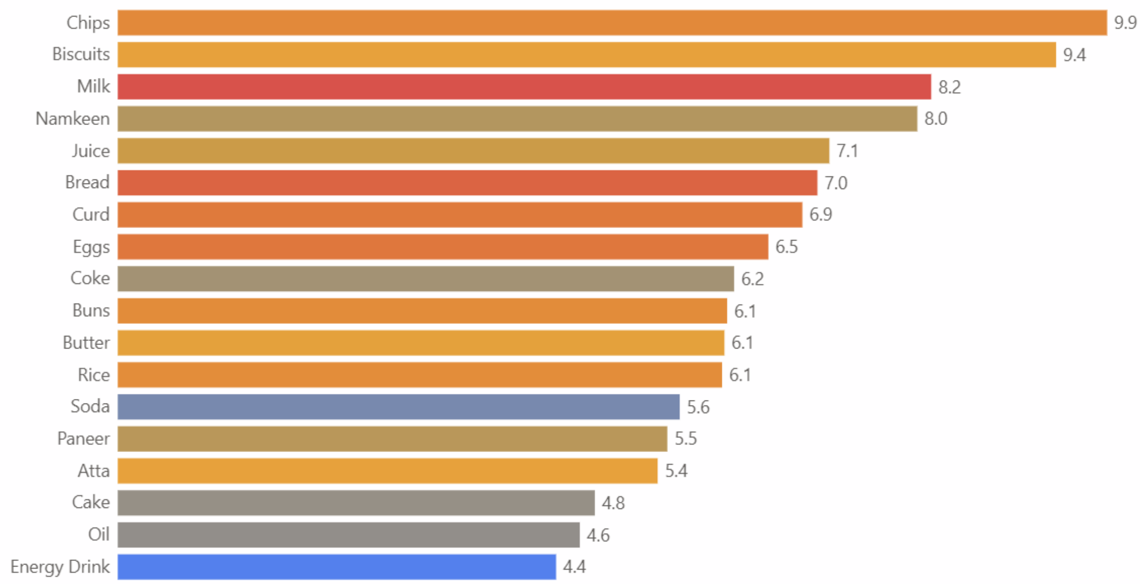


Figure 10: Forecasting accuracy and variability analysis

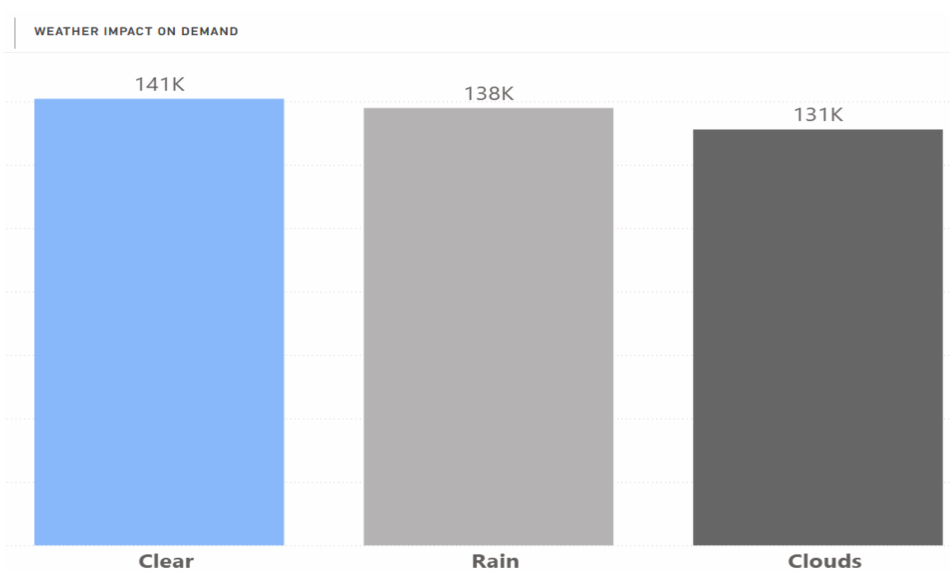



## Weather Impact & Operational Analysis

Weather conditions showed moderate influence on operational demand behavior.

Clear weather conditions generated slightly higher average demand compared to rainy and cloudy conditions.

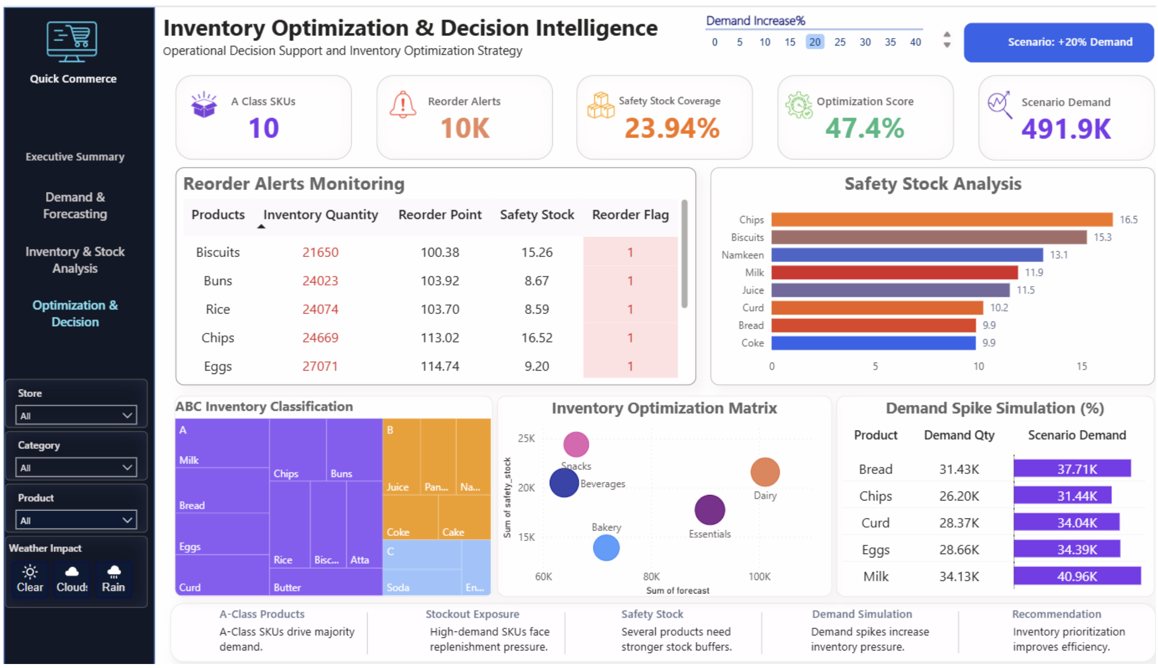
Figure 11: Weather-based operational demand analysis



# Inventory Optimization & Reorder Intelligence

Safety stock and reorder intelligence models were implemented to identify products requiring replenishment attention.

Key Findings
Reorder Alerts: 10K
Optimization Score: 47.4%
Safety Stock Coverage: 23.94%
Scenario Demand: 491.9K
Figure 12: Inventory optimization and replenishment intelligence



# Executive Dashboard Storytelling

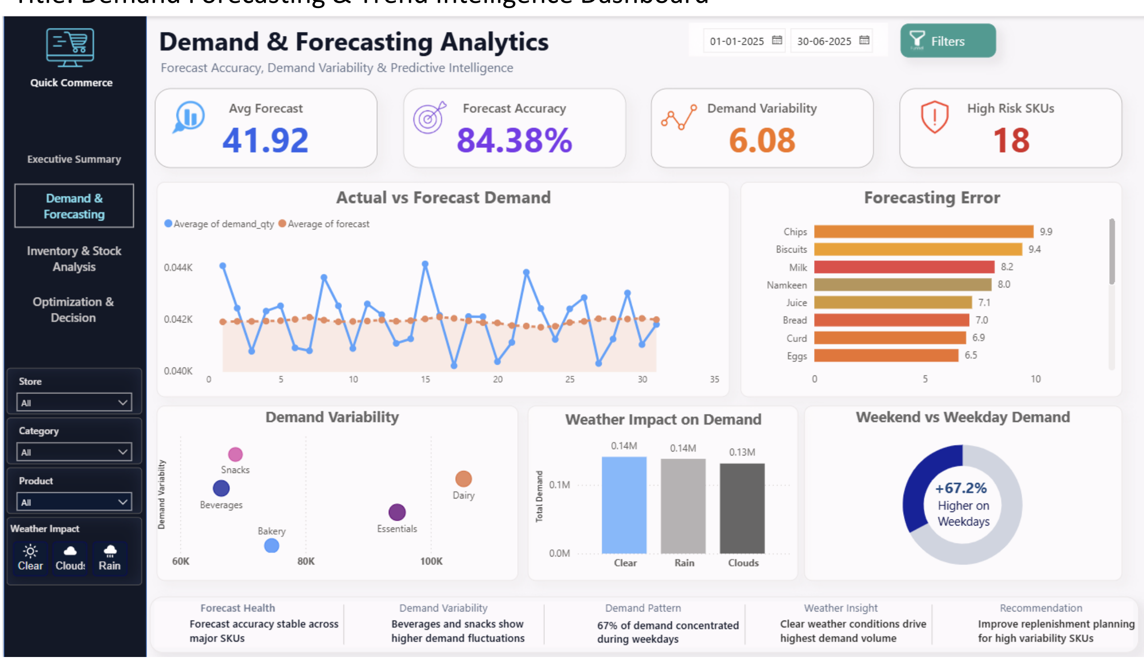
## DASHBOARD 1

Title: Operational KPI & Stockout Monitoring Dashboard


Interpretation: The dashboard provided visibility into operational demand, stockout exposure, fulfillment performance, and lost sales trends across stores and categories.

## DASHBOARD 2

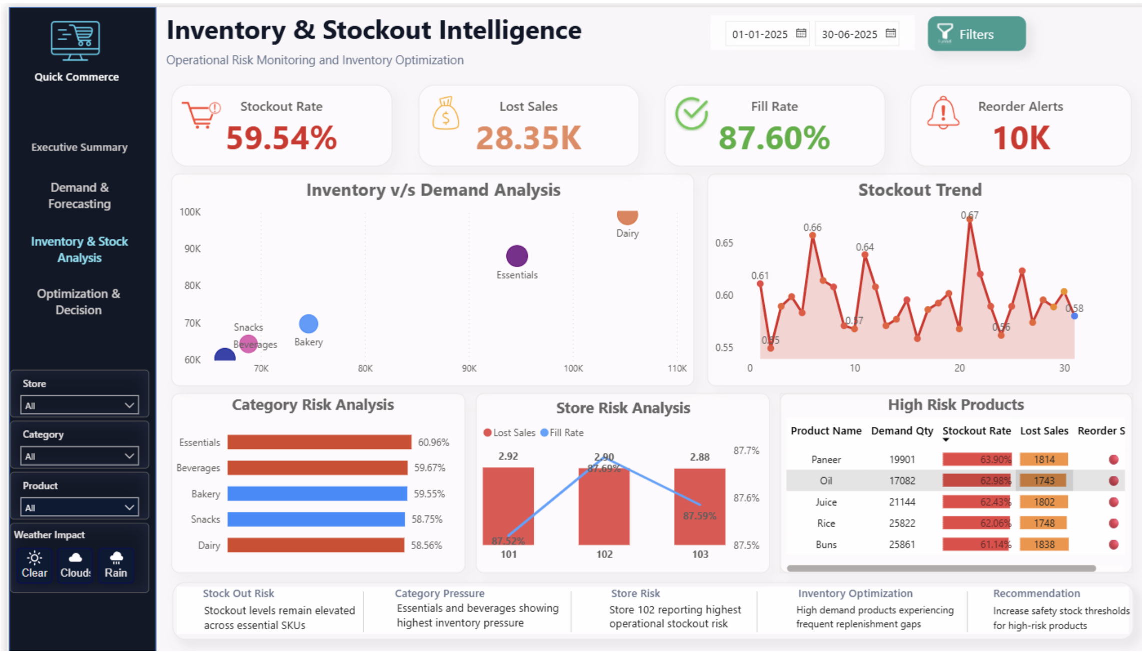
Title: Demand Forecasting & Trend Intelligence Dashboard



**Interpretation:** The forecasting dashboard highlighted demand movement, forecasting accuracy, weekend demand behavior, and operational demand variability.

## DASHBOARD 3

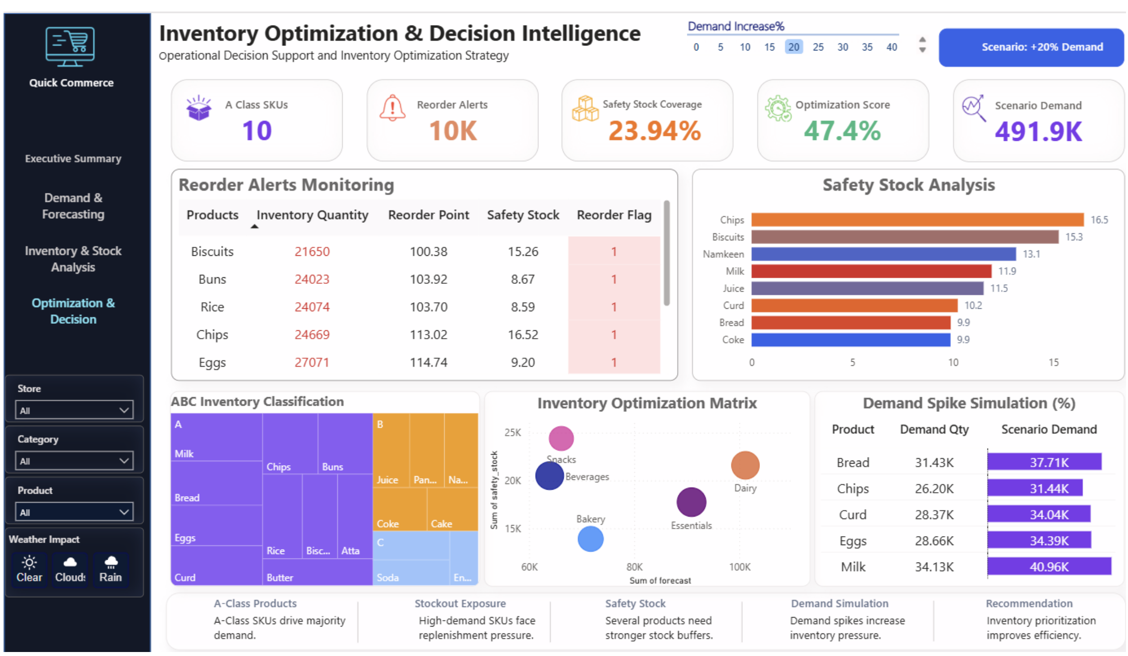
Title: Inventory Risk & Product Performance Dashboard




**Interpretation**: The dashboard identified high-risk products, category-level inventory pressure, and operational stockout exposure across stores.

## DASHBOARD 4

Title: Inventory Optimization & Decision Intelligence Dashboard



Interpretation: The optimization dashboard supported replenishment planning, safety stock monitoring, reorder intelligence, and demand spike simulation analysis.

## Final Recommendations

Inventory Recommendations
• Increase safety stock coverage for high-risk products
• Improve replenishment frequency for Essentials and Beverages
• Strengthen monitoring for fast-moving products

Forecasting Recommendations
• Implement adaptive forecasting for volatile products
• Monitor weekend demand spikes separately
• Improve category-level forecasting visibility

Operational Recommendations
• Reduce lost sales through proactive reorder planning
• Improve stock allocation efficiency across stores

## Conclusion

This project successfully transformed operational retail data into meaningful business intelligence using Python, SQL, and Power BI.

The analysis improved visibility into demand behavior, inventory performance, stockout exposure, forecasting variability, and replenishment planning.